



Deprescribing antihypertensives in older people in primary care: subgroup analysis of the MINOR randomised clinical trial

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Abstract

Background Polypharmacy is often required for older adults with hypertension, and excessive treatment is associated with a high risk of adverse reactions, including hypotension.

Aim To evaluate the deprescribing of antihypertensive medications guided by pharmacists using home blood pressure monitoring in older adults with hypotension.

Method A subgroup of older adults with signs or symptoms of hypotension, included in the MINOR clinical trial, was analysed. In the MINOR procedures, each patient was provided with a device to conduct blood pressure measurement at home for 1 week, following which a report was generated and shared with pharmacists (intervention group) or family physicians (control group). In the intervention group, a pharmacist suggested optimising pharmacotherapy; meanwhile, in the control group, a family physician alone determined the necessary treatment adjustments. Differences in mean blood pressure, the patients with symptoms/signs of hypotension, and the antihypertensive medication deprescribing between both groups were analysed.

Results Seventy-two patients were evaluated (35, control group; 37, intervention group). The intervention group showed a significant reduction in medication prescriptions (−28.6%; $P < 0.001$), especially beta-blockers (−74.2%), loop diuretics (−83.3%), and aldosterone antagonists (−80%). The mean office blood pressure in the intervention group increased (14.1 mmHg systolic and 6.9 mmHg diastolic), remaining below the target range (140/90 mmHg). The intervention group showed a significantly reduction in hypotensive symptoms than the control group (64.9% vs. 20%) ($P < 0.001$).

Conclusion The data highlight an important role for pharmacists in optimizing hypertension management in older people. Deprescribing antihypertensives can limit symptomatic hypotension.

Trial Registration Registered on ClinicalTrials.gov under number NCT04861727.

Keywords Arterial hypertension · Deprescription · Hypotension · Primary health care

Impact statements

- Hypotension is a common and dangerous adverse event in older adults receiving antihypertensive pharmacotherapy. Home blood pressure monitoring is a useful tool for identifying hypotension.
- Deprescribing antihypertensive medications can help control blood pressure and reduce polypharmacy and hypotensive symptoms in older adults.
- Pharmacists can play a role in optimizing the management of hypertension in older adults through careful deprescribing

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Introduction

Hypertension, a non-communicable and multifactorial disease, remains a major global public health concern and a primary risk factor for cardiovascular diseases (CVD) [1]. Hypertension is a common comorbidity among older people and a contributing factor to increased mortality in this age group [2, 3]. When deciding on a treatment plan for patients with this medical condition, it is important to consider the disease's stage at diagnosis and any additional risk factors. This is especially important for older populations, as there is no consensus on specific blood pressure goals [1]. Emphasising patient individuality is essential, considering not only age but also functional status, frailty, existing comorbidities, and polypharmacy [1, 4, 5].

Intrinsic characteristics in managing hypertension, such as the white coat effect, can impact clinical practice and lead to overdiagnosis and overtreatment [1, 6], consequently resulting in inadequate therapeutic decisions and increasing the risk of adverse reactions, such as hypotension [6, 7]. This is characterised by a sustained reduction with systolic blood pressure (SBP) ≤ 120 mmHg and diastolic blood pressure (DBP) ≤ 70 mmHg while on antihypertensive medication. When coupled with medication use, hypotension poses a significant concern due to the risks of falls and fractures [1, 8].

To better manage the clinical conditions of this subpopulation of patients, blood pressure monitoring strategies in the office must be reassessed. An increased number of measurements, particularly outside the office, correlates with greater predictive power for cardiovascular outcomes [1, 9]. In the routine clinical assessment conducted by pharmacists, home blood pressure monitoring (HBPM) can enhance hypertension control and prevent overtreatment in primary care settings [1, 10]. One possible alternative to mitigate issues associated with overtreatment in older adults is deprescribing, defined as the identification and supervised discontinuation, reduction, or replacement of medications, which avoids potential problems during treatment [11, 12]. Coupled with appropriate monitoring, deprescribing manages polypharmacy and prevents associated problems, including orthostatic hypotension [11, 13, 14].

Implementing interventions stemming from this process in primary care is challenging [15]. It is essential to overcome various barriers in the process of deprescribing, such as patients' uncertainty about the benefits of reducing or stopping a specific medication, professionals' uncertainty about when and how to proceed with deprescribing, clinical guidelines that encourage medication use (rather than deprescribing), and the absence of a comprehensive professional training strategy. Furthermore, it is necessary to implement health education initiatives within the community to involve patients in the process of deprescribing [11, 16–18].

Aim

The aim of the study was to evaluate the deprescribing of antihypertensive medications guided by pharmacists using home blood pressure monitoring (HBPM) in older adults with hypotension.

Ethics approval

This study was conducted in accordance with the principles of the Declaration of Helsinki and was approved by the local research ethics committee on March 17, 2021 (approval number 4,595,144).

Method

Study design

This was an analysis of a subgroup of individuals from the MINOR study, a randomised, open-label, non-blinded study with two parallel groups, assessing baseline and post-intervention (use of technology to optimise hypertension treatment in older adults in primary care) [19]. The study was conducted between June 2021 and August 2022 in a Brazilian community pharmacy. The MINOR participants were patients aged 60 years or older who had been diagnosed with hypertension and filled their prescriptions for antihypertensive medications at a public community pharmacy in Brazil. In Brazilian community pharmacies, pharmacists typically provide guidance on medication use and care to patients, especially those with chronic conditions like hypertension. They play a crucial role in optimizing drug therapy in collaboration with other healthcare professionals. [19].

In the MINOR study, in both groups, baseline office blood pressure was measured in duplicate during the initial visit and recorded as the mean following the Brazilian guidelines recommendations [1]. At the beginning of the study, all patients were interviewed, and variables of interest were collected, including the prescription of antihypertensive medications. The patients were provided with a blood pressure device to conduct HBPM for 7 days and were then randomly assigned to either the control or intervention groups. Due to the nature of the intervention and standard practices in pharmaceutical care, neither the patients nor the pharmacists were blinded. In the intervention group, the pharmacist followed up with participants after the 7-day HBPM period, conducting a clinical assessment and pharmacotherapy review informed by a guideline. This review included an analysis of the HBPM report, identification of any signs of hypotension, and providing deprescribing suggestions to the participants and their family physicians, who

considered the pharmacists' report in their care decisions. Conversely, patients in the control group received standard care: the general practitioner received an HBPM report, analysed the data, and decided on deprescribing medications. No additional pharmaceutical intervention was provided beyond the standard practice of monitoring and reporting for the control group. Office blood pressure measurements, HBPM, and verification of prescribed medications were repeated 45 days following the initial measurement and consultation with a family physician (Fig. 1). Details of the study design, eligibility criteria, and main results have been previously reported [19].

Population

The analysed subgroup comprised 72 older adults showing signs and symptoms of hypotension within a total sample of 322 MINOR patients. Signals or reports of dizziness, blurred vision, tremors, asthenia, palpitations, syncope, headache, mental confusion, and falls indicated hypotension. Additionally, hypotension was confirmed when the mean home blood pressure showed SBP ≤ 120 mmHg and DBP ≤ 70 mmHg while on antihypertensive medication [20, 21].

Outcomes

Differences in mean blood pressure, patients with symptoms/signs of hypotension, and antihypertensive medication

deprescribing in the control and intervention groups were analysed.

Statistical Analysis

The data were analysed using IBM SPSS Statistics version 27.0 (IBM SPSS Statistics, IBM Corporation, Armonk, New York, USA). The Kolmogorov–Smirnov test was used to assess the normality of the variables, and the *t*-test combined with Levene's test was used to evaluate the mean difference in blood pressure in the HBPM and the difference in the mean number of antihypertensive medications before and after intervention. Pearson's χ^2 test was utilised to determine differences between groups in the deprescribing percentage and 95% confidence intervals. Differences with *P*-values < 0.05 were considered statistically significant.

Results

We identified a total of 72 patients with blood pressure lower $< 120/70$ mmHg who reported clinical symptoms of hypotension between June 2021 and August 2022, of which 35 were in the control group and 37 in the intervention group.

No difference was observed in the mean number of antihypertensive medications used per patient or mean blood pressure (systolic and diastolic) between the groups in the baseline. The baseline mean office-based SBP and DBP were higher than those of the HBPM measurements, but the difference was less than 1 mmHg in both cases (Table 1).

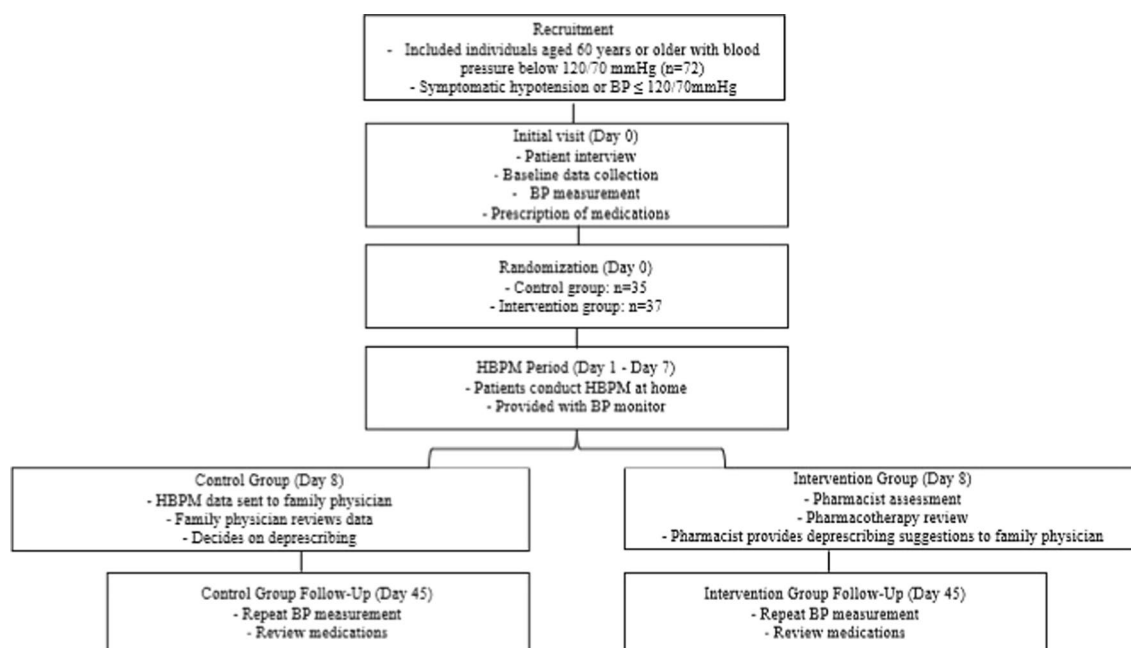


Fig. 1 Flowchart of the study population

Table 1 Baseline data of individuals who exhibited blood pressure lower than 120/70 mmHg and clinical symptoms of hypotension during the initial study assessment

Variables	Mean		t-test for equality of means			
	Control	Intervention	<i>p</i> value	Mean difference	CI (95%)	
					Lower	Upper
Age (years)	69.46	70.19	0.46	−0.73	−2.74	1.27
Time since diagnosis (years)	14.26	14.03	0.84	0.23	−2.14	2.60
Number of antihypertensives	4.14	4.08	0.70	0.06	−0.26	0.38
Systolic blood pressure at the office (mmHg)	103.11	103.27	0.89	−0.16	−2.61	2.29
Diastolic blood pressure at the office (mmHg)	70.43	71.46	0.13	−1.03	−2.39	0.32
Systolic blood pressure in HBPM (mmHg)	102.77	102.38	0.73	0.39	−1.86	2.65
Diastolic blood pressure in HBPM (mmHg)	70.54	71.05	0.38	−0.51	−1.68	0.66

Following interventions, the mean number of antihypertensive medications prescribed in the intervention group (2.95) was reduced compared to the control group (3.70) ($P < 0.001$). The intervention group showed a greater increase in the mean SBP and DBP than the control group, as confirmed by both office and HBPM measurements. In the intervention group, mean office blood pressure increased by 14.2 mmHg systolic and 7.5 mmHg diastolic ($P < 0.001$), and mean home blood pressure increased by 12.7 mmHg systolic and 5.9 mmHg diastolic ($P < 0.001$), both remaining below the target range of 140/90 mmHg (Table 2).

At the end of the follow-up period, a lower percentage of patients in the intervention group (35.1%) showed signs or symptoms of hypotension than that in the control group (80%) ($P < 0.0001$). This study identified five groups of antihypertensive medications used to treat patients: thiazide diuretics, loop diuretics, aldosterone antagonists, angiotensin AT1 receptor blockers, angiotensin-converting enzyme inhibitors, calcium channel blockers, and beta-blockers. There was a notable reduction in medications in the intervention group (−28.6%; $P = 0.001$), with an 83.3% decrease in loop diuretics compared with 27.2% in the control group ($P < 0.001$), an 83% reduction in

aldosterone antagonists compared with 47.4% in the control group ($P < 0.001$), and a 74.2% decrease in beta-blockers compared with 21.2% in the control group ($P < 0.001$).

Discussion

The results demonstrated that the intervention group, which received pharmacist reviews and recommendations, experienced a notable reduction in the number of antihypertensive medications used. Compared to the usual care group, the intervention group showed a greater increase in both systolic and diastolic blood pressure levels, both at home and in the office, while remaining within the target range. Additionally, the intervention group exhibited a more substantial reduction in symptoms of hypotension. These findings highlight the effectiveness of pharmacist-led interventions in optimising hypertension management and minimising adverse effects in older adults.

One of the main strengths of this study is its focus on a subgroup of older adults who are particularly vulnerable to the adverse effects of antihypertensive medications, thereby providing targeted insights into the benefits of deprescribing in this population. The use of HBPM allowed for a

Table 2 Data from the MINOR clinical trial subgroup after intervention

Variables	Mean		t-test for equality of means			
	Control	Intervention	<i>p</i> value	Mean difference	CI (95%)	
					Lower	Upper
Number of antihypertensives	3.70	2.95	<0.001	0.71	0.33	1.08
Systolic BP at the office (mmHg)	109.31	117.37	<0.001	−8.06	−11.15	−4.97
Diastolic BP at the office (mmHg)	73.91	78.40	<0.001	−4.49	−6.47	−2.52
Systolic BP at home (mmHg)	108.60	115.97	<0.001	−7.37	−10.32	−4.42
Diastolic BP at home (mmHg)	73.14	77.45	<0.001	−4.32	−6.11	−2.53

comprehensive assessment of blood pressure changes in a real-world setting, enhancing the validity of the findings. However, this study had some limitations. The follow-up period was relatively short, which may not capture the long-term effects of deprescribing. Additionally, the study was conducted at a single centre, limiting the generalisability of the findings to the broader population. Home measurements were conducted by the patients themselves, which may result in potential measurement errors. However, this risk was minimised through a robust training protocol performed by the responsible professional during in-person visits. Furthermore, this study is a secondary analysis of data, and the sample size was small, which may affect the strength and direct applicability of the findings.

Our results indicate that the HBPM measurements were less than 1 mmHg lower than the office readings for both SBP and DBP and, therefore, likely to be both clinically and statistically non-significant. This suggests that HBPM may not have been necessary to prompt deprescribing in these symptomatic patients, raising questions about the necessity of including HBPM as part of the intervention. However, it is acknowledged that HBPM may have been useful at follow-up to reassure both patients and their healthcare professionals that hypertension control was maintained despite deprescribing [19].

The increase in SBP and DBP measured both in-office and at home in both groups after the deprescribing process aligns with previous findings, which showed a modest rise in blood pressure (3.4 mmHg in SBP and 2.5 mmHg in DBP) [22]. Despite this increase, the new values did not indicate a heightened cardiovascular risk as they remained within the cutoff points established for older adults. The ideal SBP goals for older patients remain a topic of significant contention, with differing guidelines suggesting thresholds ranging from stringent (< 120 mmHg) to more relaxed (SBP < 140 or < 150 mmHg based on cardiovascular risk) [23].

In different clinical situations, a gradual dose reduction or even discontinuation of antihypertensive medications may be necessary, especially in cases of symptomatic hypotension or persistent reduction in systolic blood pressure below the target, either inside or outside the office [24, 25]. In the primary healthcare setting, pharmacists' deprescribing as part of collaborative medication management has been well-received by patients and other healthcare professionals [25]. Apart from a proven reduction in the number of medications [26], deprescribing inappropriate medications in patients with chronic diseases benefits patients [27]. In our study, a guideline-directed pharmacist to ensure that the patients had no other compelling indication for therapy (e.g., Heart Failure with Reduced Ejection Fraction) before recommending

deprescribing one of the antihypertensive agents. Despite having observed the positive impact of deprescribing in reducing hypotension symptoms in our study, long-term evaluation of this process remains warranted [28].

In relation to the medications, diuretics were the most commonly deprescribed in the older adults studied. When selecting antihypertensive treatments for this population, it is essential to consider the prevalence of comorbidities and specific contraindications that may affect medication choice. Diuretics are linked to hypotension and a high risk of falls in older adults, which supports their frequent discontinuation in our study. This finding highlights the importance of carefully balancing the benefits and risks of antihypertensive therapy in older adults, especially when aiming to reduce potential adverse effects such as falls, a significant concern in this population [29–31].

Future research should focus on longer follow-up periods to better understand the long-term impacts of deprescribing antihypertensive medications in older adults. It is essential to verify whether the observed increase in blood pressure poses any long-term cardiovascular risk. Exploring the impact of deprescribing on other health outcomes, such as cognitive function and quality of life, would provide a more comprehensive understanding of the benefits and risks associated with this intervention.

Conclusion

The results of this study highlight the critical role of pharmacist intervention in reducing the number of antihypertensive medications taken by older adults, thereby helping to control blood pressure and minimise hypotensive symptoms. The pharmacist's involvement was instrumental in optimising hypertension management. Furthermore, these findings underscore the importance of carefully evaluating deprescribing decisions in this group to ensure the balance between effective blood pressure control and the minimisation of adverse effects.

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Conflicts of interest The authors declare that they have no conflict of interest.

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